

BIOPHARMACEUTICS AND CLINICAL PHARMACOKINETICS



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Introduction

- Drugs are combined with a number of inert substances and transformed into a convenient DF that can be administered by a suitable route.
- Earlier, it was believed that the therapeutic response to a drug is an attribute of its intrinsic pharmacologic activity.
- But today, it is understood that the dose response r/ship obtained after drug administration by different route are not the same.

➤ Variations are also observed:

- Same drug is administered as d/t DF
- Physicochemical properties of the drug
- The excipients present in the DF
- The method of formulation
- Similar DFs produced by d/t manufacturers

—→ A new and separate discipline called *biopharmaceutics* has therefore been developed to account for all such factors that influence the therapeutic effectiveness of a drug.

➤ **Biopharmaceutics** is defined as the study of factors influencing the *rate* and *extent* of drug that reaches the systemic circulation and the use of this information to optimize the therapeutic efficacy of the drug products.

- *Knowledge of biopharmaceutics is essential for rational drug product design and optimum drug therapy*
- **Absorption**: is the process of a movement of a drug from its site of administration to systemic circulation.
- The concentration of drug in plasma and hence the onset of action, and the intensity and duration of response depend upon the bioavailability of drug from its DF.
- **Bioavailability???**
- *Any alteration in the drug's BA is reflected in its pharmacologic effects.*
- Other processes that play a role in the therapeutic activity of a drug are distribution and elimination.

- ***Distribution***: is the dispersion of substances throughout the fluids and tissues of the body i.e. the movement of drug between one compartment and the other.
- The **intensity** and the **duration of action** depend largely upon the effective concentration and the time period for which this concentration is maintained at the site of action which in turn depend upon the ***elimination*** processes.
- ***Elimination*** is defined as the process that tends to remove the drug from the body and terminate its action. Elimination occurs by two processes:
 - ***Metabolism***: which inactivates the drug
 - ***Excretion***: responsible for the exit of drug/metabolites from the body

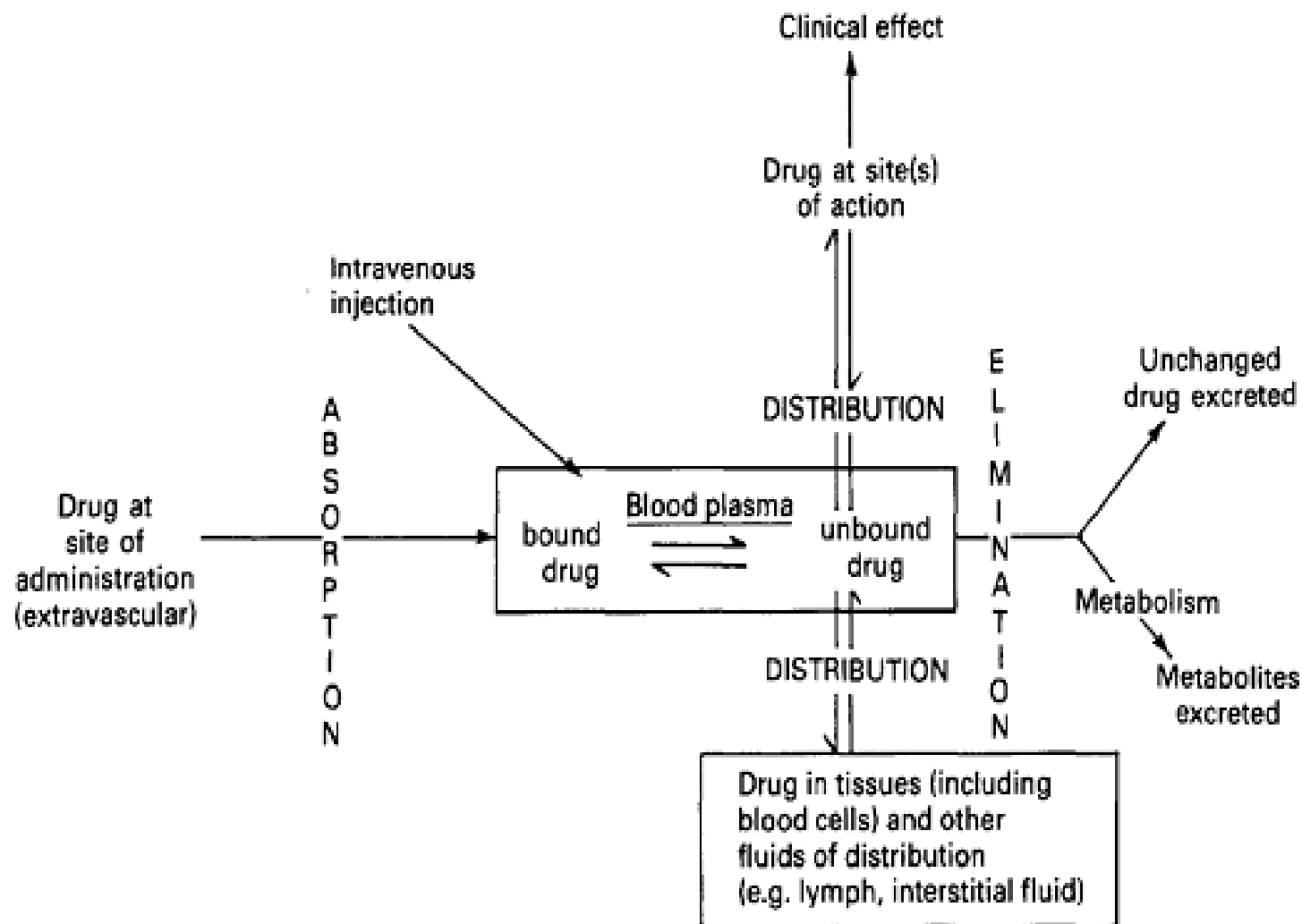
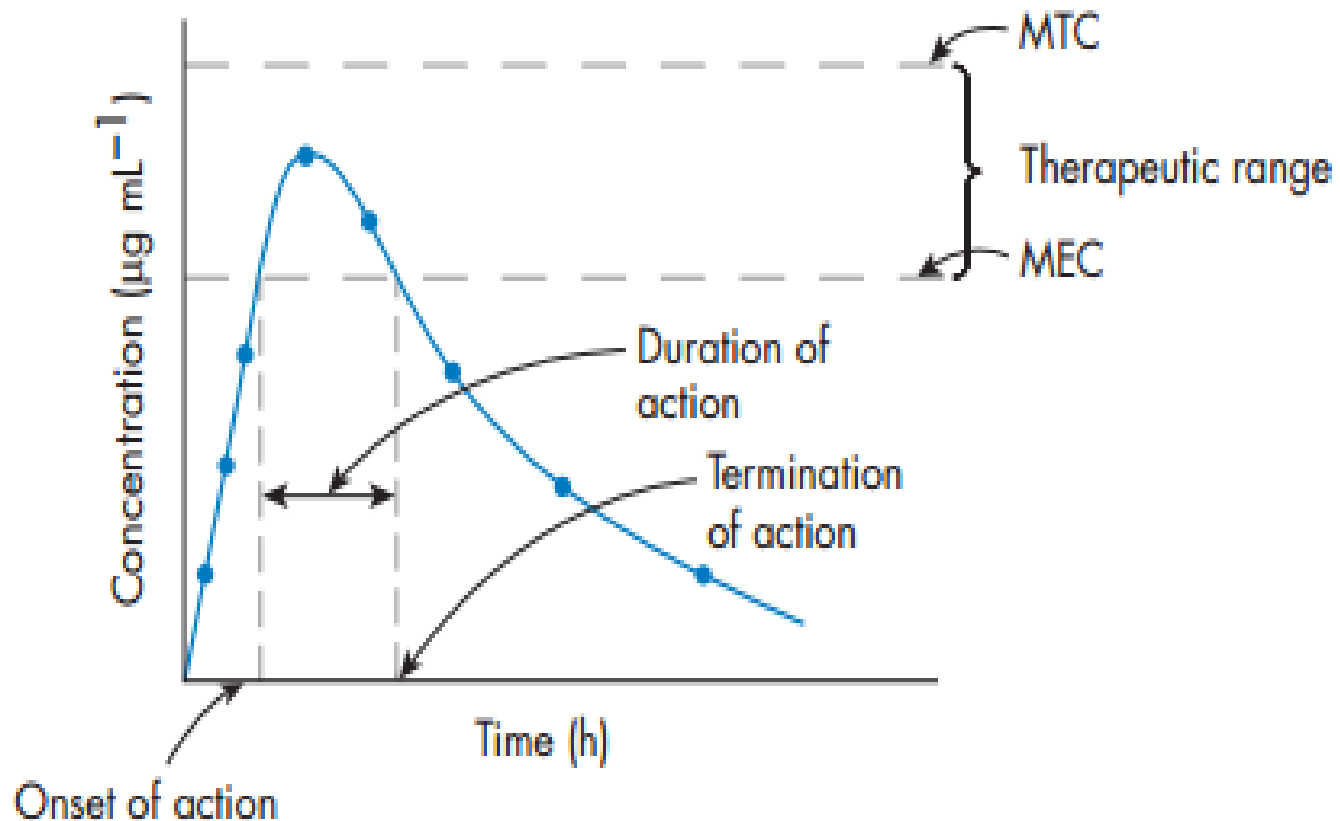


Figure: Factors that influence concentration of drug at its site(s) of action

- In order to administer drugs optimally, knowledge is needed not only of the mechanisms of drug **A**bsorption, **D**istribution, **M**etabolism and **E**xcretion (**ADME**) but also of the rate (kinetics) at which they occur i.e. *pharmacokinetics*.
- ***Pharmacokinetics***: is the study of time course of drug ADME and their relationship with its therapeutic effects of the drug
- ***Clinical pharmacokinetics***: the use of pharmacokinetic principles in optimizing the drug dosage to suit individual patient needs and achieving maximum therapeutic utility.
- Drug administration and therapy can now be conveniently divided into 4 phases or processes:-

1. *The Pharmaceutical Phase*: It is concerned with the formulation of an effective DF of the drug for administration by a suitable route.
2. *The Pharmacokinetic Phase*: It is concerned with the ADME of drugs as elicited by the plasma drug concentration-time profile. It is the sum of all the processes inflicted by the body on the drug
3. *The Pharmacodynamic Phase*: It is concerned with the biochemical and physiological effects of the drug and its mechanism of action. It is characterized by the concentration of drug at the site of action and its relation to the magnitude of effects observed.
4. *The Therapeutic Phase*: It is concerned with the translation of pharmacological effect into clinical benefit.

- Onset time
- Duration of action
- MEC, MTC
- Therapeutic range



Drug Product Performance Parameters:

- 1- **Minimum effective concentration (MEC):** The minimum concentration of drug needed at the receptors to produce the desired pharmacologic effect.
- 2- **Minimum toxic concentration (MTC):** The drug concentration needed to just produce a toxic effect.
- 3- **Onset time:** The time required for the drug to reach the MEC.
- 4- **Duration of action:** The difference between the onset time and the time for the drug to decline back to the MEC

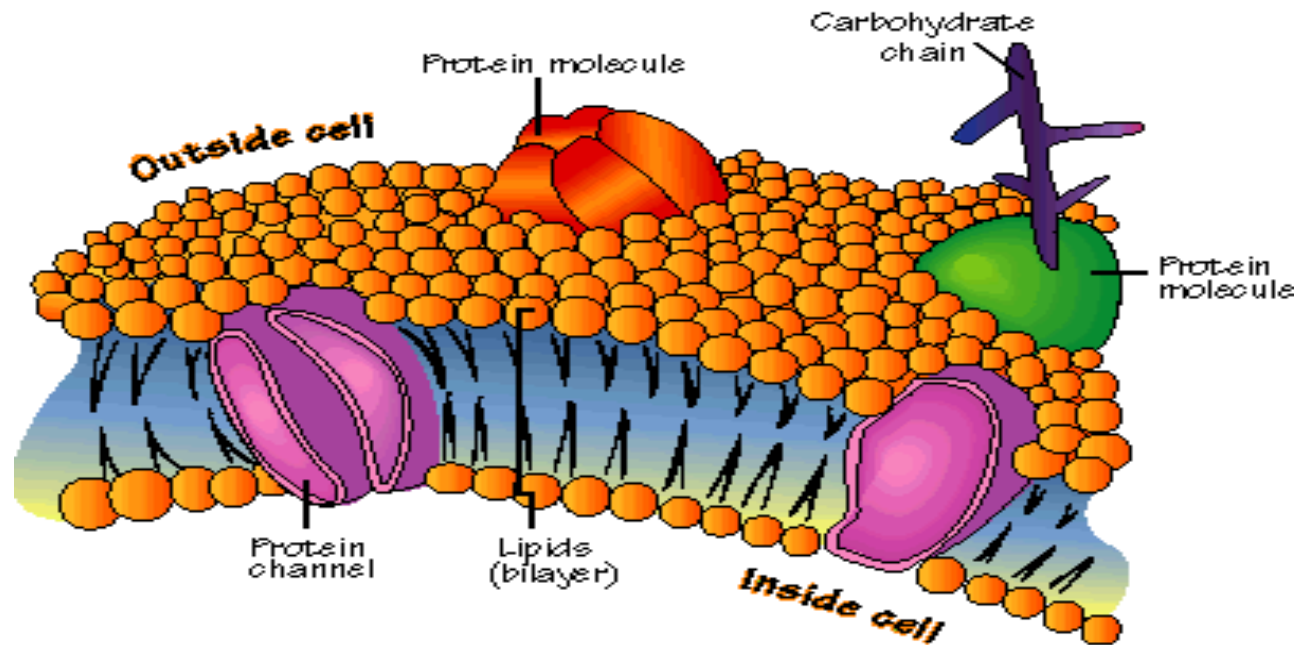
*The knowledge and concepts of **BP** & **PK** thus have an integral role in the design and development of new drugs and their DFs and improvement of therapeutic efficacy of existing drugs*

PHYSIOLOGY OF MEMBRANE

- ❑ The oral route of drug administration is the most common for systemically acting drugs and therefore, more emphasis will be given to GI absorption of drugs.
- ❑ Before proceeding to absorption aspects, a brief description of cell membrane structure and physiology is necessary.
- ❑ Functionally, cell membranes are semipermeable partitions that act as selective barriers to the passage of molecules.
 - **Controls traffic in & out of the cell**
 - **selectively permeable**

PHYSIOLOGY OF MEMBRANE...

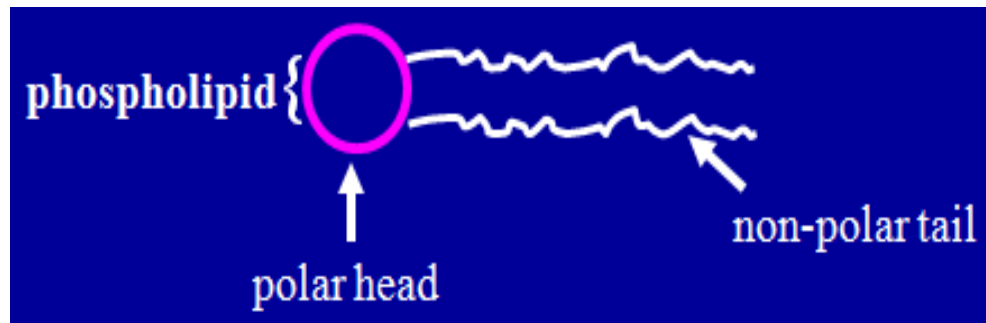
- ❑ Cell membranes are composed primarily of phospholipids in the form of a bilayer interdispersed with carbohydrates and protein groups.



- ❑ Cell membranes are generally thin, approximately 70 to 100 Å in thickness.

PHYSIOLOGY OF MEMBRANE...

- ❑ The phospholipids make up a bilayer. It contains hydrophilic and hydrophobic molecules.



- ❑ Because of the lipoid nature of the cell membrane, it is highly permeable to lipid soluble substances.
- ❑ It also contains small aqueous channels or pores

PHYSIOLOGY OF MEMBRANE...

Semipermeable lipoidal sieve

- Allow rapid transit of some materials and impede others
- Permeable to nutrients: amino acids, sugars, fatty acids, ...
- Allows passage of lipid-soluble molecules across it
- Allow passage of water and small hydrophilic molecules through its numerous aqueous pores
- Highly charged and large molecules don't

PHYSIOLOGY OF MEMBRANE...

- ❑ Number of transporter proteins or carrier molecules transport materials back and forth across it.
- ❑ For systemic drug absorption, the drug must cross cellular membranes.
- ❑ Once in the plasma, the drug may have to cross biological membranes to reach the site of action.
- ❑ Therefore, biological membranes potentially pose a significant barrier to drug delivery.